Re-analysis Summary Report for Peer-evaluation

Paper ID: **McDevitt_JournPoliEco_2014_yQeR**

Re-analyst's ID: **anl1q**

Your task

As a peer-evaluator, for this paper you are asked to judge whether the analyst's applied analytical choices for Task 1 and Task 2 are acceptable. By acceptable, we mean that the analysis pipeline is within the variations that could be considered appropriate by the scientific community in addressing the given analytical task. In addition, for Task 1, we ask you to judge whether the reported conclusion adequately follows from the results of the analysis; whether the co-analyst's self-categorization of the result is adequate. Although, you are not required, it would be appreciated, if you could execute the accompanying analysis code to check any mismatch between the results of the analysis and the reported results.

For both tasks, you should provide your evaluation via this survey: https://forms.gle/EAREaC6JjN2PWFfR6

The re-analyst's task

The re-analyst was instructed to assess the following claim of the original authors:
The average plumbing firm whose name begins with A or a number receives… more service complaints than other firms …(p. 909.)

The corresponding article

Here you can find the original article: https://osf.io/6fusy/?view_only=c874fcf097644e82a320679d4641163a and the original data: https://osf.io/z7mfu/?view_only=277b7d18bb24436eb5cfca69a07eb07a

They are provided only as context of the re-analysis.

In Task 1, the analyst was asked to conduct the analysis without any restrictions. Here, the analyst reported the following steps and results of the analysis:
I did not perform any preprocessing steps to the provided dataset (i.e., "final_data.dta"). First, I checked the distribution of the dependent variable "complaints_2008". This inspection reveals a very high number of zeros and a low number of companies with a very high number of complaints. Thus, outliers and zero-inflation could be an issue for the statistical analyses. Second, to enable an analysis that is robust against outliers, I dichotomized the dependent variable and calculated a Chi-square test for the resulting 2x2 matrix. This test was significant, indicating that firms with an "A"-name have a higher likelihood of receiving at least one complaint. Third, I compared a Poisson, a negative binomial, and a zero-inflated generalized linear model in their capacity to model the data because these types of models are recommended for count data. It turns out that a negative binomial model shows the best fit. It is, however, important to note that all three modeling approaches point to a significant effect of a plumbing firm's name on the number of complaints the firm receives (i.e., when the name begins with an "A", the number of complaints is increased). Fourth, I included control variables (i.e., multiple_names, on_google, ad_spend_k, firm_age, Chicago, and emp_size) in a negative binomial model, which does not change the significance of the core effect. Finally, I also estimated a t-test to replicate the results reported in Table 2 of the paper (i.e., showing that the mean

number of complaints a firm receives is higher for firms whose name begins with "A" than for firms with other names). This test is also significant. In sum, the key effect of interest is significant for all employed modeling approaches.

The analyst's conclusion: There is a significant effect of a plumbing firm's name on the number of complaints the firm received such that firms whose name begins with "A" have a higher likelihood of receiving a complaint.

The analyst's categorization of the result:
The results show evidence for the relationship/effect as described in the claim provided in your task

In Task 2, the analysts received the following instruction to conduct the analysis:
Use only the data of Illinois Plumbing Firms in your model. Do not restrict your sample to those firms that serve the metro Chicago area. Do not include the reviews received via yelp.com, Angie's List, and Consumer's Checkbook into your analysis. You should use the overall complaints number instead of complaints per employee.

Here, the analyst reported the following steps and results of the analysis:
I did not perform any preprocessing steps to the provided dataset (i.e., "final_data.dta"). To the best of my knowledge (based on some cross-validations), this dataset fulfills the criteria that it only includes data of Illinois Plumbing Firms, that it does not include reviews from yelp.com, Angie's List, and Consumer's Checkbook, and that it includes the total number of complaints instead of complaints per employee. I did not restrict the analysis to firms that serve the metro Chicago area. Hence, all requirements for task 2 are fulfilled. Because I already based my analyses of the first task on these requirements, my analyses for task 1 and task 2 are identical, and I will focus my subsequent reporting on the most informative statistical model.

Due to the very high number of zeros in the dependent variable "complaints_2008", a negative binomial model shows the best fit to the data. Due to the correlative nature of the dataset, I consider it important to include relevant control variables (i.e., multiple_names, on_google, ad_spend_k, firm_age, Chicago, and emp_size) when testing the key effect of interest. The analysis reveals, as hypothesized, that firms whose name begins with "A" receive a higher number of complaints than firms with other names ($b = 0.68$, $z = 3.121$, $p = .002$). All included control variables but the firm age ($p = .198$) have significant and positive effects ($ps < .014$).

Analysis language/software/code (optional to check):
Task 1: R
Task 2: R
Analysis codes: https://osf.io/yp98z/